Checking delivery routes against traffic delays: statistical screening versus travel-time padding in vehicle routing

Abstract

Background and objective. Delivery companies promise customers a time window, but they plan routes on average travel times, and one delay can make a driver miss several windows. The usual protection is to plan with slower speeds, padding every trip. This study asks whether a statistical check on an ordinary planner's routes is a better protection than padding. **Methods.** A standard solver produces candidate plans with extra internal slack, and each candidate is simulated on 1000 days with time-of-day congestion and correlated random delays. A plan is accepted only if a one-sided 95% Wilson lower bound on the chance that at least 95% of customers are served on time reaches 0.95, and the accepted plan is then tested on 5000 new days. A cap on the slack keeps every customer reachable. The study used 53 synthetic instances with 20 to 200 customers, a padding sweep from 0 to 50%, and three confirmatory studies whose hypotheses were published before they were run. **Results.** Screening was 8.3–21.4 percentage points more reliable than 20% padding, although not significantly at 200 customers. No padding level matched it over all instances: the best, 30%, pooled 66.7% against 86.6%, and heavier padding often found no plan at all. Seven of eight prespecified primary hypotheses were confirmed. **Conclusions.** A simple statistical screen can make an existing planner's routes more reliable without changing the planner. The evidence comes from simulated cities and an assumed delay model, so it still needs testing on real traffic data.

Keywords: vehicle routing; time windows; travel-time uncertainty; simulation; Wilson interval; operations research

1 Introduction

A courier company that promises to deliver between 10:00 and 11:00 has to plan which driver visits which customers, and in what order, so that every visit falls inside its window. This is the vehicle routing problem with time windows (VRPTW), one of the most studied problems in operations research[1,2]. Planning software usually solves it with one fixed table of travel times. Real travel times are not fixed. Roads are slower in the morning rush, and random events such as an accident or a long wait at a door add delay. Because a driver visits customers one after another, a delay early in the day pushes every later visit towards the end of its window, and a plan that looked comfortable can fail.

The simplest protection, and a common one in practice, is to plan with a slower speed than expected. If every trip is assumed to take 20% longer, the plan has room to absorb delays. The drawback is that padding protects every trip equally, quiet off-peak trips as much as rush-hour

ones, so the plan uses extra distance and sometimes extra vehicles whether or not a route is at risk. Research methods that model uncertainty directly, through chance constraints or robust optimization[3–5], spend that protection more carefully, but they need a solver built for the purpose, which most companies do not have.

This study tests a middle path. The company keeps its ordinary solver, but asks it for several candidate plans, each planned against internal deadlines brought forward by a few minutes. Customers never see these internal deadlines, so the promised windows do not change. Each candidate is then simulated on many random days, and a statistical test decides whether it is reliable enough. The best plan that passes is used. Its reliability is reported from a second, independent set of simulated days, so the number reported is not the one used to choose the plan.

The study makes three contributions. First, it describes this screening procedure and proves that a simple cap on the internal slack prevents a failure in which a customer becomes impossible to reach in time. Second, it compares the procedure with padding at levels from 0 to 50%, measuring both reliability and cost, and shows that no single padding level does as well across instances. Third, it repeats the main comparisons in three confirmatory studies whose hypotheses were fixed and published before the experiments were run, using two different routing solvers.

2 Methods

2.1 The routing problem

A depot and n customers are given. Customer i has a demand, a service time of 10 minutes and a window $[l_i, u_i]$ in which service must start. Each vehicle carries at most 1000 units and works at most 600 minutes. A plan assigns customers to vehicles, orders the visits and fixes a departure time for each vehicle. Plans are compared on four quantities: total distance, total driver time, number of vehicles, and slack deficit, which measures how close planned arrivals come to the end of their windows. The planner minimizes a weighted sum of these four quantities, each scaled to a comparable range, with the weights $(0.5693, 0.2643, 0.1055, 0.0609)$ fixed before any experiment. Two solvers were used: OR-Tools from Google[6], a widely used routing library, and the hybrid genetic search (HGS) of the PyVRP package[7,8], which finds shorter routes. The full mathematical formulation is given in the project archive[9].

2.2 Travel times and random delays

Each day runs from 06:00 with a free-flow speed of 40 km/h. Between 07:00 and 09:00 speeds drop by a factor of 1.2, and between 06:00 and 07:00 they are slightly higher. Travel times are computed by integrating the speed over the time of the trip, so a vehicle that leaves later never arrives earlier, a property known as first-in-first-out[10]. On a simulated day, the time of each trip is multiplied by a random factor $\exp(\varepsilon)$ with

$$\varepsilon_{ij} = \sigma\left(\sqrt{\rho}\,\xi_0 + \sqrt{1-\rho}\,\xi_{ij}\right), \tag{1}$$

where ξ_0 is a standard normal variable shared by every trip on that day (a bad-traffic day slows everyone) and ξ_{ij} is an independent standard normal variable for the trip from i to j. The main setting uses $\sigma = 0.2$ and $\rho = 0.5$. The factor has median one, so on a typical day trips take about

their planned time. A fixed set of simulated days is called a *bank*; every plan for an instance is tested on the same banks, so comparisons are paired.

2.3 What counts as reliable

A day counts as a success for a plan if at least 95% of its customers are served on time. The target is that a plan succeeds on at least 95% of days. Asking for *every* customer to be on time would be unrealistic for large instances: if each of 200 customers were independently on time with probability 0.99, all 200 would be on time on only about 13% of days, because $0.99^{200} \approx 0.134$.

A plan's success rate on 1000 simulated days is only an estimate, so the test uses a lower confidence bound. If a plan succeeds on a share $\hat{p}$ of N days, the one-sided Wilson lower bound[11] at 95% confidence is

$$L(\hat{p}) = \frac{\hat{p} + \dfrac{z^2}{2N} - z\sqrt{\dfrac{\hat{p}(1-\hat{p})}{N} + \dfrac{z^2}{4N^2}}}{1 + \dfrac{z^2}{N}}, \qquad z = 1.645. \tag{2}$$

The Wilson bound was chosen because the simpler normal-approximation interval is unreliable when the true probability is close to one[12], which is exactly the region of interest here. With $N = 1000$ and $\hat{p} = 0.966$, for example, $L \approx 0.955$, so the plan passes; with $\hat{p} = 0.955$, $L \approx 0.943$ and it fails.

2.4 The screening procedure

The procedure has four steps.

1. For each buffer $\beta \in \{0, 5, 10, \ldots, 60\}$ minutes, ask the solver for a plan in which every customer's internal deadline is $u_i - \beta$ (never earlier than l_i). This gives up to 13 candidate plans.

2. Simulate every candidate on a screening bank of 1000 days and compute its lower bound (2).

3. If at least one candidate has $L \geq 0.95$, select the passing candidate with the best weighted score. Otherwise return a fallback plan fixed in advance: the plan with no buffer for the OR-Tools procedures, and the 20% padded plan for the HGS procedures.

4. Simulate the selected plan once on a separate validation bank of 5000 days and report its success rate.

Every simulated arrival is always judged against the window u_i promised to the customer, never against the internal deadline. Because the validation bank is not used to choose the plan, the reported reliability is an honest out-of-sample estimate. The average validated success rate over all instances, including those where the fallback was used, is called *pooled service*.

2.5 Keeping every customer reachable

Bringing deadlines forward can backfire. If customer i is far from the depot, an internal deadline $u_i - \beta$ can be earlier than the time it takes to drive there directly, and then no plan can exist at that buffer. Let $e_i = \max\{l_i, \bar{t}_{0i}\}$ be the earliest time at which service at i could start after a direct trip from the depot, where $\bar{t}_{0i}$ is the planned driving time. The *capped* procedure uses the internal deadline

$$u_i^{\mathrm{cap}}(\beta) = \min\big\{u_i,\ \max\{l_i,\ u_i - \beta,\ e_i\}\big\}. \tag{3}$$

Lemma 1. *For every customer i and every buffer $\beta \geq 0$: (a) $u_i^{\mathrm{cap}}(\beta) \leq u_i$, so a plan that meets the capped deadlines also meets the promised windows; (b) if customer i can be reached in time by a direct trip, that is if $e_i \leq u_i$, then $e_i \leq u_i^{\mathrm{cap}}(\beta)$, so the capped deadline never makes a reachable customer unreachable; and (c) if $\beta \leq u_i - e_i$, the cap has no effect and $u_i^{\mathrm{cap}}(\beta) = \max\{l_i, u_i - \beta\}$.*

Proof. (a) holds because $u_i^{\mathrm{cap}}(\beta)$ is a minimum that includes u_i. For (b), the inner maximum is at least e_i, and $u_i \geq e_i$ by assumption, so the minimum of the two is at least e_i. For (c), $\beta \leq u_i - e_i$ gives $u_i - \beta \geq e_i$, so the inner maximum equals $\max\{l_i, u_i - \beta\}$, which is at most u_i because $l_i \leq u_i$ and $\beta \geq 0$; the outer minimum therefore leaves it unchanged. □

The lemma guarantees reachability one customer at a time. It does not guarantee that a single plan can serve all customers in time, because capacity, shift length and the order of visits still matter, and it does not guarantee that the target is met. A separate shortest-path test was used to prove, where possible, that a contracted model had no feasible plan, so that solver failures could be told apart from true infeasibility.

2.6 Comparisons

The main comparison is with padding: plan for the shortest distance at a slower speed, $40/(1+p)$ km/h, so that every trip is padded by p percent. The earlier studies used $p = 20\%$; this paper adds a sweep over $p \in \{0, 10, 20, 30, 40, 50\}\%$, with the same solver budget, to compare cost at equal reliability. Padded plans are not screened; they are used as an operator would use them. The *original* procedure screens contracted candidates without the cap, and the *capped* procedure uses (3). A third comparison, described in the Results, used a budgeted robust optimization model[4,13], a method that plans against a set of worst-case travel times.

2.7 Instances and statistics

Instances place a depot at the centre of a 100 km by 100 km square and customers uniformly at random, with demands from 10 to 49 units and windows from 90 to 149 minutes wide. The development study used 53 instances: 20 with 20 customers, 15 with 50, 10 with 100 and 8 with 200. Each solver call had 5 seconds for OR-Tools or 10,000 iterations for HGS. Because all configurations are tested on the same simulated days, differences are computed instance by instance, and intervals across instances come from a paired bootstrap with 20,000 resamples. A configuration that finds no plan scores zero on that instance, so failures count against it.

Table 1. Paired differences in service probability (percentage points) on the 46 development instances where 20% padding found a plan. SC denotes screened conservative speed with its prespecified fallback.

n	S minus C	P minus C	SC minus C
20	−11.0 [−16.1, −6.0]	21.4 [9.6, 31.3]	28.2 [21.4, 34.3]
50	−13.1 [−17.4, −8.9]	17.9 [2.8, 32.0]	11.8 [2.7, 22.2]
100	−16.2 [−20.6, −11.7]	16.0 [4.8, 24.8]	11.2 [2.8, 19.4]
200	−12.9 [−18.5, −5.7]	8.3 [−5.1, 20.0]	5.7 [0.0, 12.9]

2.8 Confirmatory studies

The development study was exploratory: its results were used to design the method. To check that the findings were not a product of that tuning, three confirmatory studies were run on new instances. Before each one, its hypotheses, tests, decision rules and instance seeds were written down and published with a cryptographic fingerprint of the code[14–16], so the analysis could not be adjusted after seeing the results. Study 1 tested the OR-Tools procedures on 53 new instances; study 2 used 53 more instances and the HGS solver; and study 3 compared the procedures with the robust model on the 90 earlier confirmatory instances with up to 100 customers (part A), and tested a new fallback rule on 53 further instances (part B). Each study, and each part of study 3, had two primary hypotheses, judged with 97.5% intervals so that the chance of a false confirmation stays below 5% for each pair.

3 Results

3.1 Screening beats 20% padding

Table 1 compares the procedures with 20% padding on the 46 development instances where padding found a plan. The original screening procedure was 8.3–21.4 percentage points more reliable, depending on size. The interval at 200 customers includes zero, so the gain is clear only up to 100 customers. Using the weighted objective without any screening was *worse* than padding at every size, which shows that the gain comes from testing candidates, not from the objective alone. The advantage grew with the amount of random delay: for independent delays ($\rho = 0$), the difference rose from -3.1 points at $\sigma = 0.1$, where padding was slightly ahead, to 30.5 points at $\sigma = 0.3$ (Fig. 1).

3.2 Padding at other levels

A fixed 20% pad may simply be the wrong amount, so Table 2 and Fig. 2 repeat the comparison with padding from 0 to 50%. Over all instances, no padding level came close to the capped screening procedure. The best level, 30%, pooled 66.7% against 86.6%, a difference of -19.9 points [-28.8, -11.7]. More padding did not keep helping, because heavy padding makes many instances impossible to plan: the padded planner found a plan on 34 of the 53 instances at 40% and on only 26 at 50%.

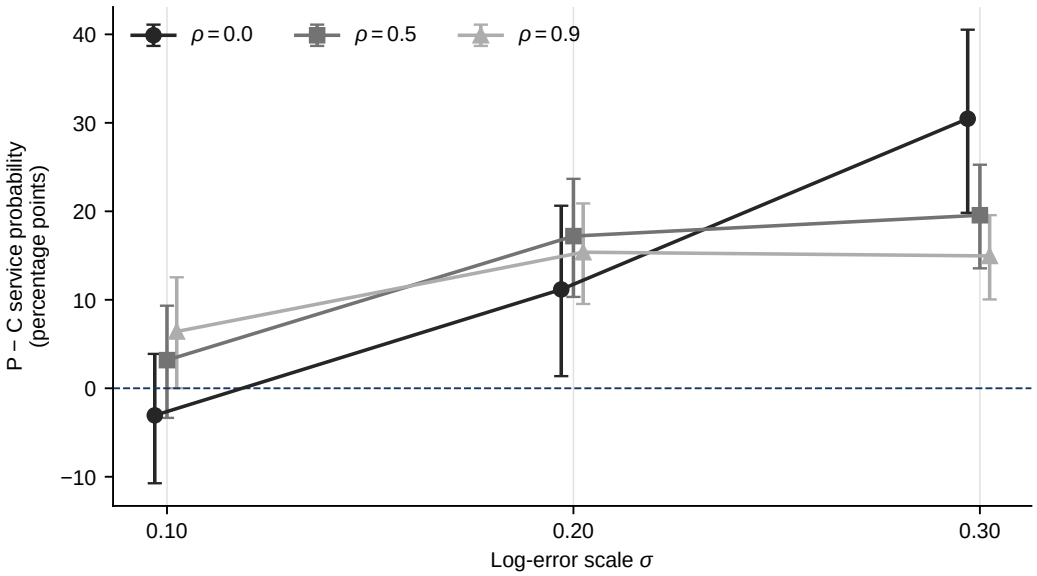


Figure 1. Difference in validated service probability between the original screening procedure (P) and 20% padding (C) on the 46 development instances, as the random delay σ and its correlation ρ vary. Plans are fixed; error bars are 95% bootstrap intervals over instances.

Where heavy padding did find a plan, that plan was reliable. On those 26 instances, 50% padding served 96.1% against 87.9% for screening, 8.2 points more [1.2, 16.1], and used 0.77 more vehicles per instance. These were the easier instances, however, and on the other 27 the padded planner returned nothing. A planner who fixes one padding level therefore has to choose between unreliable plans at low levels and missing plans at high ones. Screening avoids that choice, because it picks the buffer for each instance separately and the cap keeps every customer reachable. At 30% padding, where the padded planner did best overall, screening drove 3.1% more distance than the padded plans on the instances where both had a plan, so its extra reliability costs only a few percent of distance.

3.3 The cap removes most failures

On 21 development instances the original procedure found no plan that passed the test, and on 17 of them the shortest-path test proved that some customer had become unreachable at a large buffer. Capping (Table 3) raised the number of instances with a passing plan from 32 to 42, and pooled service by 7.7 [3.5, 12.5] points. At 100 and 200 customers the capped procedure found a passing plan on every instance and pooled 97.5–97.6%. It used 2.5% more distance and 0.43 more vehicles on average than the original procedure. Of the 42 plans that passed screening, 40 also kept a lower bound of at least 0.95 on the 5000 validation days, which shows that the screen rarely admits a plan that later disappoints.

Pooled service stayed below the 95% target because of the remaining instances, all with 20 or 50 customers, where no candidate passed and a fallback was used. Plans that passed averaged 97.4% validated service, while the 11 fallback plans averaged 45.2%.

Table 2. Padding every trip by a fixed percentage on the 53 development instances (exploratory). *Plans*: instances on which the padded planner found a plan. *All*: pooled service over all instances, an instance without a plan scoring zero. The next three columns use only the instances with a padded plan: the padded plans' service, the capped screening procedure's service on the same instances, and the difference (padding minus screening, percentage points, 95% paired bootstrap interval). Distance (%) and vehicles are also padding minus screening on those instances. Over all instances the capped procedure pooled 86.6%.

		All	Instances with a padded plan				
Padding	Plans	Pooled	Padded	Screened	Difference	Distance	Vehicles
0%	53/53	21.9	21.9	86.6	−64.7 [−70.2, −58.8]	−7.1	−0.89
10%	51/53	43.7	45.4	88.4	−42.9 [−48.0, −37.4]	−6.2	−0.59
20%	46/53	56.8	65.4	89.4	−23.9 [−29.2, −18.2]	−4.8	−0.22
30%	43/53	66.7	82.2	90.3	−8.1 [−13.1, −2.4]	−3.1	0.21
40%	34/53	58.8	91.6	88.5	3.1 [−2.8, 10.0]	−1.0	0.47
50%	26/53	47.1	96.1	87.9	8.2 [1.2, 16.1]	0.9	0.77

Table 3. Development study, all 53 instances. For each procedure: instances admitted by screening out of all instances of that size, with the number whose lower bound also reached 0.95 on the validation bank in parentheses, and pooled validated service probability (%) with a 95% bootstrap interval, fallback plans included.

	Original procedure		Screened conservative		Capped procedure	
n	Admitted	Pooled	Admitted	Pooled	Admitted	Pooled
20	14/20 (14)	81.6 [70.2, 91.9]	15/20 (15)	87.7 [79.0, 94.7]	14/20 (14)	81.6 [70.3, 92.0]
50	8/15 (8)	72.7 [59.2, 86.4]	4/15 (4)	67.5 [58.2, 77.3]	10/15 (10)	80.1 [66.9, 90.8]
100	6/10 (5)	81.3 [68.8, 93.3]	4/10 (4)	77.5 [67.2, 88.2]	10/10 (9)	97.5 [96.7, 98.2]
200	4/8 (3)	80.6 [68.7, 92.3]	2/8 (2)	78.3 [71.0, 86.5]	8/8 (7)	97.6 [96.9, 98.2]

3.4 Confirmatory studies

Table 4 lists the eight primary hypotheses. Seven were confirmed. In study 1, screening beat 20% padding by 15.7 points on new instances, and capping added 11.7 points. In study 2, with the stronger HGS solver, capped screening beat padding by 28.0 points, so the gain does not depend on a weak solver. The hypothesis that was not confirmed (G2) concerned a change to the fallback: returning the padded plan when nothing passes did not measurably help the capped OR-Tools procedure, because that procedure rarely needed a fallback. Study 3 part B tested a different rule, returning the candidate that came closest to passing, and it raised pooled service by 7.3 and 4.2 points for the two solvers, to 94.4% and 94.9%.

3.5 Comparison with robust optimization

A budgeted robust model plans against worst-case travel times on a limited number of trips. Without a reachability correction it failed on 16 of 45 development instances at its most protective setting, for the same reason the uncapped procedure failed. With the analogous correction and the same screen, the robust plans were more reliable than both screened procedures in study 3 part A (by 6.7 and

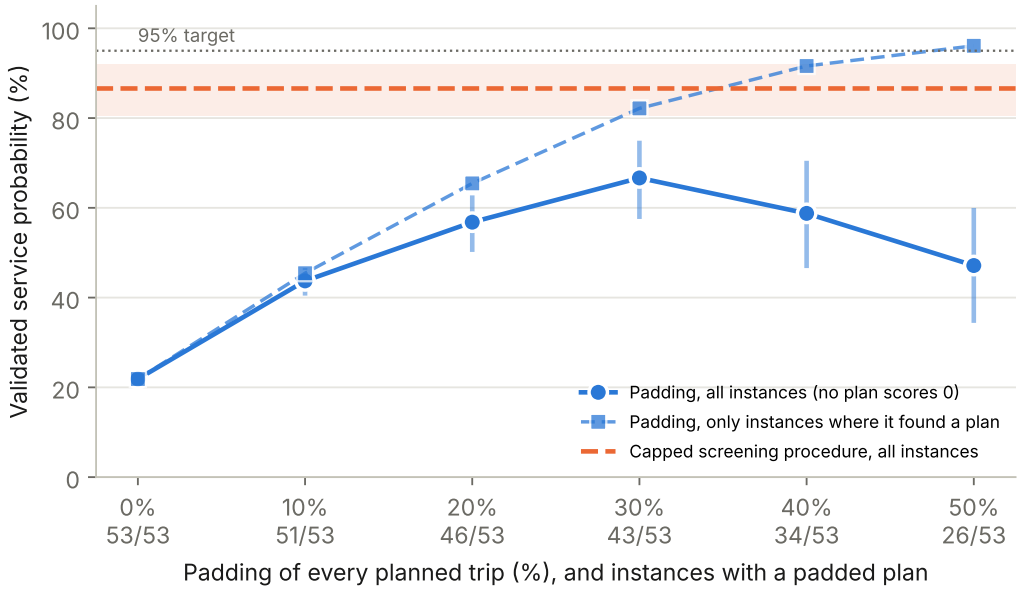


Figure 2. Validated service probability against the padding applied to every trip, on the 53 development instances. The solid line counts an instance without a padded plan as zero; the dashed line uses only the instances where a padded plan exists (counts under the axis). The orange line and band show the capped screening procedure over all instances with its 95% bootstrap interval.

6.7 points; Table 4), but they used 0.42 more vehicles per instance than the HGS procedure. When they were limited to no more vehicles than the comparison plan, their advantage was no longer statistically detectable. Screening is therefore not the most reliable planner tested; its strength is that it works with any existing solver.

4 Discussion

The results support a practical point. A company that already uses a routing solver can make its plans more reliable than a fixed 20% padding without changing the solver, by asking it for a few buffered variants and keeping the best one that passes a simulation test. The test is cheap: screening a candidate took 3 to 24 ms, against several seconds to generate it. Each accepted plan also comes with a stated, tested reliability, which padding does not provide. The padding sweep shows why choosing the buffer per instance matters: a fixed pad is either too small to protect most routes or so large that many instances can no longer be planned. The reachability cap is a one-line change, and Lemma 1 shows why it is safe; the same idea also rescued the robust model.

The study has clear limits. The cities are synthetic squares and the delay model is assumed, not estimated from real traffic data, so the percentages are not forecasts for a real fleet. The confirmatory studies show that the findings repeat on new instances from the same generator; they do not show that they hold in a different city. The advantage also depends on the planner's objective: with equal weights on the four quantities it fell to 6.3 points, with an interval that included zero. The Wilson bound holds for each candidate separately, not for all 13 at once; a stricter Bonferroni version removed all 11 admitted plans that later fell short, while 1 new selection fell short, at a cost of at

Table 4. The eight primary hypotheses of the three confirmatory studies. Each was fixed and publicly timestamped before its instances were solved. Estimates are differences in validated service probability (percentage points) with 97.5% paired bootstrap intervals over instances; a hypothesis is confirmed when its interval lies above zero.

Study		Contrast	Estimate [interval]	Decision
1	H1	Original procedure minus 20% padding (50 instances)	15.7 [7.5, 23.6]	confirmed
1	H2	Capped minus original procedure (53 instances)	11.7 [5.9, 17.9]	confirmed
2	G1	Capped procedure with HGS minus 20% padding with HGS (49 instances)	28.0 [23.0, 32.8]	confirmed
2	G2	Conservative minus slack-aware fallback, capped OR-Tools procedure	0.4 [−0.7, 2.1]	not confirmed
3A	R1	Screened robust model minus capped OR-Tools procedure (90 instances)	6.7 [3.0, 11.1]	confirmed
3A	R2	Screened robust model minus capped HGS procedure (45 instances)	6.7 [2.2, 12.2]	confirmed
3B	F1	Best-screened minus default fallback, capped OR-Tools procedure	7.3 [2.3, 13.2]	confirmed
3B	F2	Best-screened minus default fallback, capped HGS procedure	4.2 [1.2, 8.0]	confirmed

most 2 admissions per study. Finally, vehicles always leave at their planned times, and breakdowns and same-day replanning are not modelled.

Pooled service stayed just below 95% because of the few instances with no passing plan. Better candidates for those instances, for example from a solver that models uncertainty directly, are the clearest next step, together with a test on real travel-time data.

5 Conclusion

On simulated delivery problems with 20 to 200 customers, screening a family of buffered plans from an ordinary solver gave more reliable service than padding every trip by 20%, with two different solvers; pooled over all instances it also beat every fixed padding level from 0 to 50%, and three confirmatory studies with published hypotheses confirmed seven of their eight primary hypotheses. A simple cap on the buffer, proved safe in Lemma 1, removed a failure that also affects robust optimization. The method's main appeal is that it adds a measurable reliability guarantee to the tools companies already use.

Acknowledgments

[Removed for review.]

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